



# HISTORY & EVOLUTION OF CYBERSECURITY

Q From Early Computing to Modern Digital Defense Q

Presented BY:

Shane Lyza Amado  
Ma. Nilda Apura  
Khea Mae Cadalin





# INTRODUCTION

- Cybersecurity plays a vital role in protecting modern digital systems.
- As technology evolves, cyber threats also continue to grow.
- Understanding the history of cybersecurity helps us appreciate the need for strong digital protection.
- This presentation discusses the origins, growth, and future of cybersecurity.





# DEFINITION OF CYBERSECURITY

03/13

Cybersecurity refers to the protection of:

- Computers
- Networks
- Servers
- Mobile devices
- Data

Its main goal is to prevent:

- Unauthorized access
- Cyber attacks
- Data theft
- System damage

**Cybersecurity ensures data confidentiality, integrity, and availability.**





- Early computers were used mainly by governments and research institutions.
- Systems were isolated and secured physically using locked rooms and restricted access.
- The introduction of multi-user systems led to the creation of:
  - Passwords
  - User access controls
- These were the foundations of modern cybersecurity.

# EARLY BEGINNINGS (1940S-1960S)





# RISE OF VIRUSES & MALWARE [1970S-1980S]

05/13



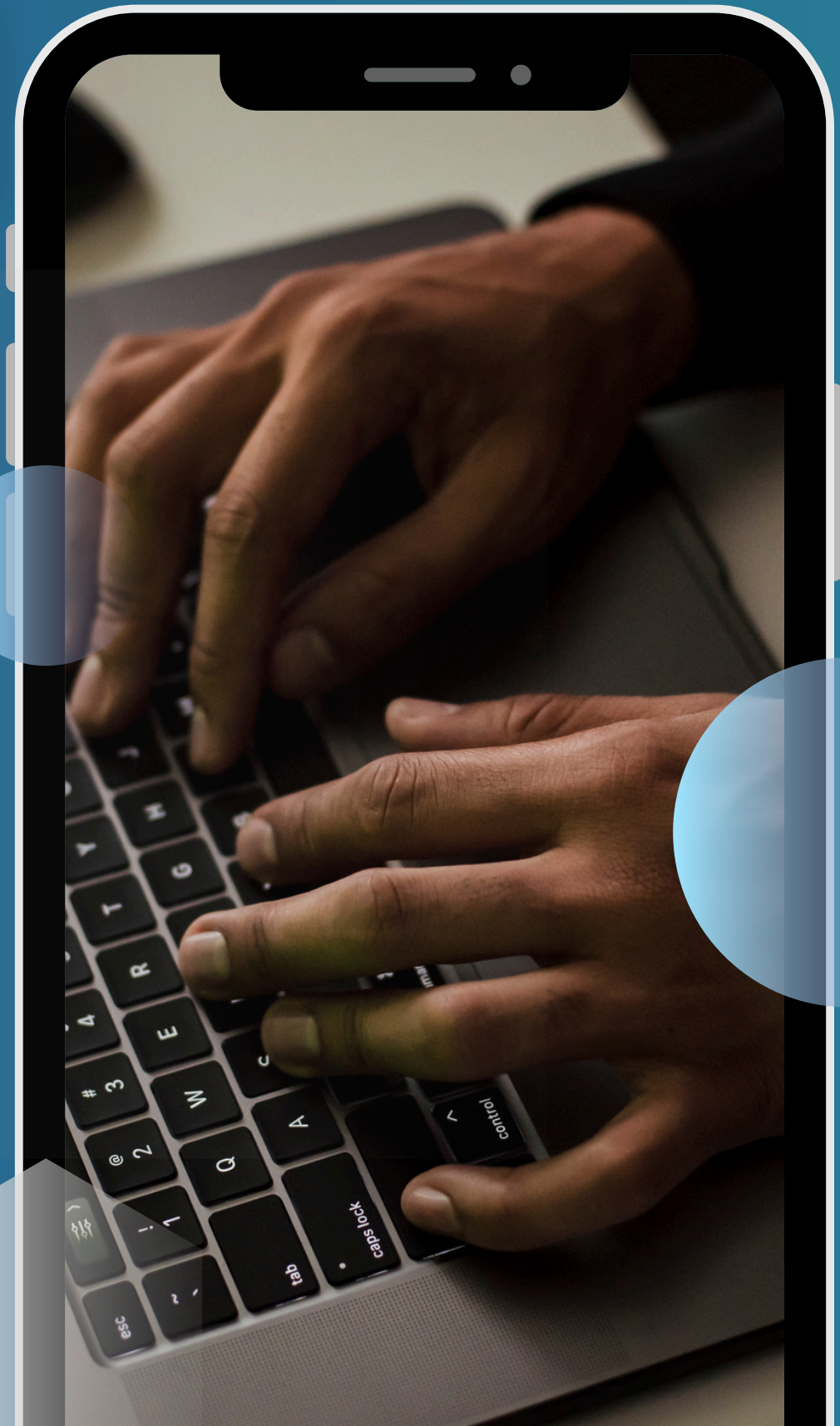
- In 1971, the first known computer virus called **Creeper** appeared.
- In 1986, **Brain Virus** became the first PC virus.
- These threats led to the development of:
  - \* Antivirus programs
  - \* Basic firewalls
- This period marked the beginning of malware defense.





# INTERNET EXPANSION (1990S)

- The rapid growth of the internet increased cybersecurity risks.
- New cyber threats emerged such as:
  - Hackers
  - Worms
  - Email viruses
- Security innovations included:
  - Network firewalls
  - Secure Sockets Layer (SSL) encryption
  - Intrusion Detection Systems (IDS)





- Cyber attacks became more organized and financially motivated.
- Common threats included:
  - Phishing
  - Spyware
  - Identity theft
  - Botnets
- Governments and organizations created:
  - Cybersecurity laws
  - International security standards
  - Security Operations Centers (SOC)



# CYBERCRIME ERA (2000S)





# MODERN CYBERSECURITY (2010S-PRESENT)



- Advanced cyber threats include:
  - Ransomware
  - Advanced Persistent Threats (APT)
  - Zero-day attacks
- New security approaches:
  - Artificial Intelligence (AI)
  - Cloud security
  - Zero Trust Architecture
  - Multi-factor authentication (MFA)





### Cybersecurity protects:

- Financial systems
- Healthcare records
- Government databases
- Educational platforms

### Prevents:

- Financial loss
- Identity theft
- Privacy violations

Ensures trust and safety in digital environments.

# IMPORTANCE OF CYBERSECURITY TODAY





# FUTURE OF CYBERSECURITY

- Artificial Intelligence-based threat detection
- Quantum cryptography
- Biometric security systems
- Automated defense mechanisms
- Strong focus on privacy and ethical security





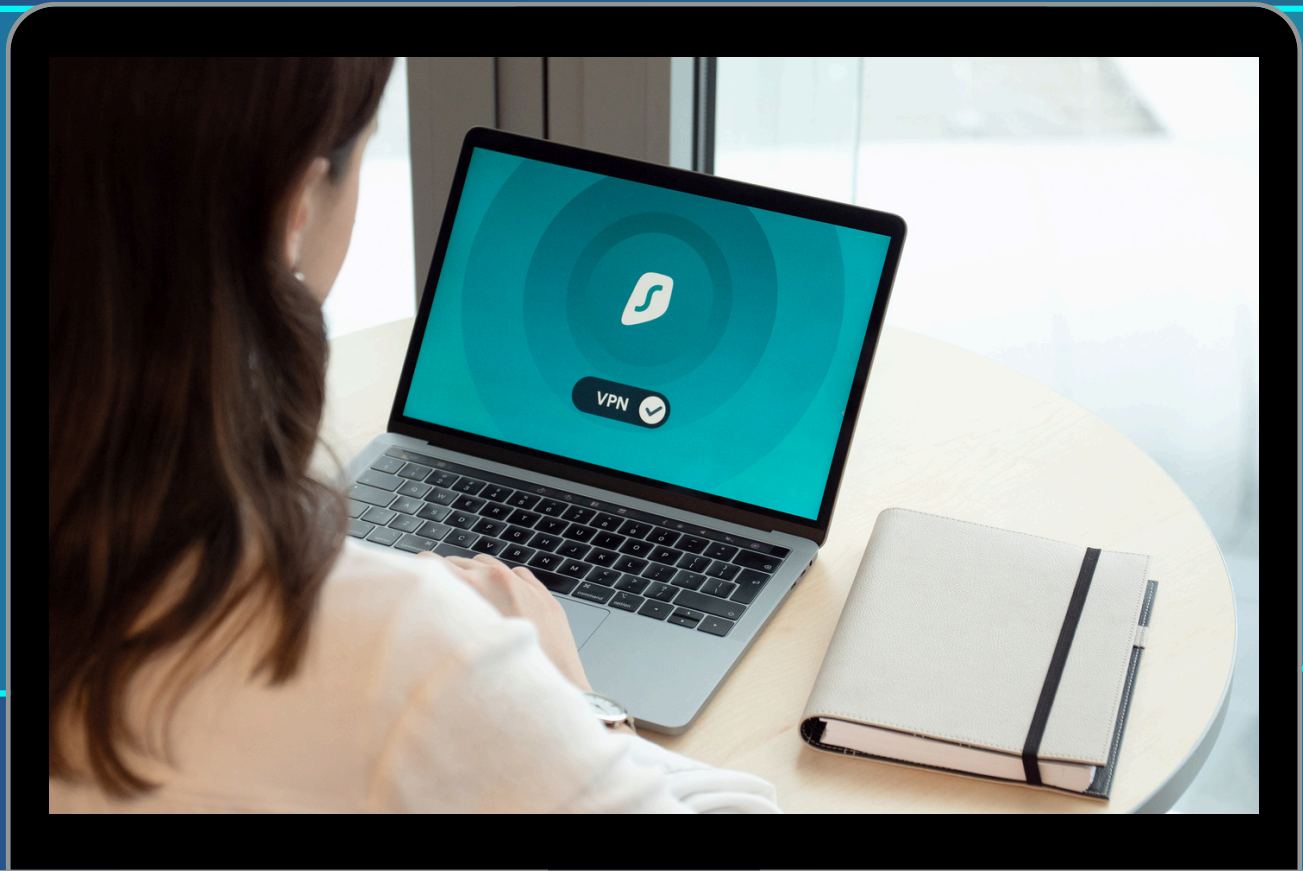
# TIMELINE SUMMARY

## Era

- 1940s–1960s
- 1970s–1980s
- 1990s
- 2000s
- 2010s–Present

## Focus

- Physical and access security
- Virus and malware protection
- Network and internet security
- Cybercrime prevention
- AI and cloud-based security





# CONCLUSION

Cybersecurity has evolved alongside technological advancements.

Cyber threats continue to become more complex and dangerous.

Strong cybersecurity practices are essential to protect individuals, organizations, and nations.

Everyone plays a role in maintaining digital safety.





 **THANK YOU!**

